

Erdős Problem #1032

EP-1032

L1 - TRACTABLE

OPEN

Graph Theory

ORIGINAL PROBLEM STATEMENT

We say that a graph is 4-chromatic critical if it has chromatic number 4, and removing any edge decreases the chromatic number to 3.

Is there, for arbitrarily large n , a 4-chromatic critical graph on n vertices with minimum degree $\gg n$?